COUNTABLE FAMILY CATCHING ALL FIBRE ISOMORPHISMS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Countable finite type pairs. lemma-countable-finite-type (same chapter, candidate proof is a stub, not carded). Enumerates pairs over the countable subfield.
- Limits descent. limits-lemma-descend-finite-presentation, limits-proposition-characterize-locally-finite-presentation (cross-chapter, named). Descend X, Y, S and the isomorphism h_s to a finitely generated subalgebra.

Source reference. Original lemma 0G0U: sources/stacks-equiv/pinned-original-source.tex, source lines 4309–4309. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZM: focused-seeds/geometry/S9-eqv-countable-isos/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZC: focused-seeds/geometry/S9-eqv-countable-isos/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let K be an algebraically closed field. Let S be a finite type scheme over K. Let $X \to S$ and $Y \to S$ be finite type morphisms. There exists a countable set I and for $i \in I$ a pair $(S_i \to S, h_i)$ with the following properties*

1. *$S_i \to S$ is a morphism of finite type, set $X_i = X \times_S S_i$ and $Y_i = Y \times_S S_i$,*
2. *$h_i : X_i \to Y_i$ is an isomorphism over S_i, and*
3. *for any closed point $s \in S(K)$ if $X_s \cong Y_s$ over $K = \kappa(s)$ then s is in the image of $S_i \to S$ for some i.*

Proof. The field K is the filtered union of its countable subfields. Dually, $\mathrm{Spec}(K)$ is the cofiltered limit of the spectra of the countable subfields of K. Hence Limits, Lemma 01ZM guarantees that we can find a countable subfield k and morphisms $X_0 \to S_0$ and $Y_0 \to S_0$ of schemes of finite type over k such that $X \to S$ and $Y \to S$ are the base changes of these.

By Lemma 0G0U there is a countable set I and pairs $(S_{0,i} \to S_0, h_{0,i})$ such that

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(1) $S_{0,i} \to S_0$ is a morphism of finite type, set $X_{0,i} = X_0 \times_{S_0} S_{0,i}$ and $Y_{0,i} = Y_0 \times_{S_0} S_{0,i}$,
(2) $h_{0,i} : X_{0,i} \to Y_{0,i}$ is an isomorphism over $S_{0,i}$.

such that every pair $(T \to S_0, h_T)$ with $T \to S_0$ of finite type and $h_T : X_0 \times_{S_0} T \to Y_0 \times_{S_0} T$ an isomorphism is isomorphic to one of these. Denote $(S_i \to S, h_i)$ the base change of $(S_{0,i} \to S_0, h_{0,i})$ by $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$. We claim this works.

Let $s \in S(K)$ and let $h_s : X_s \to Y_s$ be an isomorphism over $K = \kappa(s)$. We can write K as the filtered union of its finitely generated k-subalgebras. Hence by Limits, Proposition 01ZC and Lemma 01ZM we can find such a finitely generated k-subalgebra $K \supset A \supset k$ such that

(1) there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(A) \\ {\scriptstyle s}\downarrow & & \downarrow{\scriptstyle s'} \\ S & \longrightarrow & S_0 \end{array}$$

for some morphism $s' : \mathrm{Spec}(A) \to S_0$ over k,
(2) h_s is the base change of an isomorphism $h_{s'} : X_0 \times_{S_0, s'} \mathrm{Spec}(A) \to X_0 \times_{S_0, s'} \mathrm{Spec}(A)$ over A.

Of course, then $(s' : \mathrm{Spec}(A) \to S_0, h_{s'})$ is isomorphic to the pair $(S_{0,i} \to S_0, h_{0,i})$ for some $i \in I$. This concludes the proof because the commutative diagram in (1) shows that s is in the image of the base change of s' to $\mathrm{Spec}(K)$. $\square$